

Your Decisions Are Exploitable - The Mathematics of Adversarial Choice

In 2020 a team at the University of Melbourne published a paper in PNAS that should have restructured how security professionals think about human vulnerability. Almost nobody in the defense community read it. The paper is still mostly cited by behavioral economists.

Dezfouli, Nock, and Dayan. "Adversarial vulnerabilities of human decision-making." The thesis: human decision-making follows computational patterns that an adversarial agent can model, predict, and exploit in real time. Consistently. At the process level of cognition itself, beneath social engineering, beneath pretexting, beneath anything the target is conscious of.

The Experiment

They built adversarial agents using recurrent neural networks and ran them in competitive games against human participants. The agents observed each human's decision history, constructed a model of their sequential choice patterns, and selected counter-strategies optimized against that specific person's tendencies.

Adversarial agents achieved significantly higher win rates than game-theoretic optimal strategies. Playing the mathematically "perfect" strategy was less effective than simply modeling the human and exploiting their regularities. Participants were more exploitable than a coin flip.

The critical result: exploitation persisted even when participants knew it was happening. They could not correct the patterns through effort or awareness. The regularities exist below conscious access. Cognitive architecture does not respond to willpower.

Implications for Cognitive Security

The security industry treats humans as the "weakest link" in vague, unfalsifiable terms. Phishing training. Awareness campaigns. Dezfouli's result is far more specific and far more damaging: human cognition has a formal vulnerability surface. It can be modeled. It can be systematically attacked. With math. At scale.

If an adversarial agent can model how someone updates their strategy in a simple game, the same architecture extends to modeling belief updates in information environments. Every piece of content consumed is a move in a sequential game. Response patterns are learnable. A human social engineer reads one target at a time. An adversarial decision model processes thousands simultaneously, learning individual patterns, optimizing interventions per-person.

The targets cannot patch themselves. Knowing about the vulnerability does not close it. Dezfouli showed this empirically.

Formal Structure

The adversarial advantage comes from a basic asymmetry. Humans use bounded cognitive resources to make decisions, producing statistical regularities. The agent uses unbounded computational resources to detect those regularities and compute counter-strategies.

Dezfouli formalized this as sequence prediction. Observe a history of human choices. Build a generative model of the decision process. Select the action that maximizes expected reward given the predicted next choice. Simple architecture. The exploitation rate was statistically significant across participants with varying levels of strategic sophistication. Smart people were exploitable too, in different but equally modelable ways.

Defensive Requirements

If human decision-making is formally exploitable, cognitive defense requires more than awareness training.

Decision randomization. Injecting genuine noise into choice processes to break pattern detectability. Computationally expensive for humans and psychologically uncomfortable. People resist acting randomly. It feels wrong. Which is exactly why their non-random patterns are so exploitable.

Adversarial auditing. Running these same models against your own population to identify who is most exploitable and on which dimensions. Before someone else does.

Structural defenses. Designing information environments that make sequential pattern exploitation harder regardless of individual cognitive discipline.

Continuous model invalidation. A fixed defensive strategy is learnable. Defense must be non-stationary, which is architecturally expensive and organizationally painful.

None of these are standard practice. The distance between what Dezfouli proved in 2020 and what any organization actually implements is the operational space where cognitive warfare lives.

Every person reading this has exploitable decision patterns. They cannot be removed through training or discipline. They are structural features of sequential information processing in human cognition. Most threat models do not account for adversarial exploitation of cognitive architecture at all.

Dezfouli, A., Nock, R., & Dayan, P. (2020). Adversarial vulnerabilities of human decision-making. *Proceedings of the National Academy of Sciences*, 117(46), 29221-29228. DOI: 10.1073/pnas.2016921117

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